Judge vs. React: dissociating perception from timing in VLM reflex control

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Abstract

A vision-language-model (VLM) agent that must time a single action in a first-person scene can fail two ways that existing game benchmarks cannot pull apart: it can misjudge the scene (perception), or judge correctly but act too late (timing). We build a minimal, parametric first-person pit-jump task that reduces an episode to one correct decision, and use it to dissociate the two axes. We show (i) a **same-decision flip** — the *identical* correct decision clears the pit when the world is paused and fails in real time; (ii) a **quantitative timing law** — success collapses at a sharp threshold in the dimensionless ratio r = latency/action-window, located near $r \approx 1$ (logistic fit $r^\star = 1.17$ on the oracle apparatus), with latency varied *independently of judgment quality*; (iii) **action-repeat does not rescue it**; (iv) the boundary is a **law, not a quirk of one task** — a second reflex task in a different modality (rotational aiming) collapses onto the same $r \approx 1$ transition ($r^\star = 0.98$); and (v) a **real model traces the same boundary** — a closed-loop Sonnet agent, swept through the transition by injected delay with its own frame-by-frame judgment in the loop, collapses at $r^\star = 1.05$, and past the boundary the *only* surviving episodes are ones that misjudged early, their perception error cancelled by the delay. Measuring five real vision models on the task, **all land past $r^\star$** — hosted frontier models at $r \approx$ 4–6, local open VLMs at $r \approx$ 31–47 — so none can act inside the window in real time, yet the same decisions succeed when the world is paused. The framing implies that pausable/turn-based deployment converts a slow-but-accurate VLM from failing to viable, and that "VLMs are too slow to play games" is partly an artifact of the time regime they are benchmarked in.

1 Introduction

When a vision-language model plays a real-time game and walks into a pit, the post-mortem is ambiguous. Did it fail to *see* that a jump was needed (perception), or did it see correctly and *act too late* (timing)? Aggregate game benchmarks — score, win rate, levels cleared — fold these two failures into one number, so "the VLM is bad at this game" can mean either, and usually means some unknown mix. The two have opposite fixes: a perception failure wants a better model; a timing failure wants a slower clock. Conflating them is why "VLMs are too slow to play games" is repeated as though it were a single fact.

Our move is isolation. We reduce an episode to *one* correct decision — a first-person pit the player auto-walks toward, where the only choice is *when* to jump — so that on any given run exactly one of the two failures can occur, and we can say which. The apparatus is deterministic and headless, and it injects a control delay D (a decision made at tick t is applied at $t+D$), which lets us vary *lateness* completely independently of *judgment quality*: a perfect-judgment oracle still fails if its action lands too late.

That isolation yields five results, presented most striking first:

- **S2 (dissociation):** the same correct decision succeeds when the world is paused and fails in real time, isolating latency from perception on a single decision.
- **S1 (timing law):** success collapses at a sharp threshold in r = latency/window near 1, with latency varied independently of judgment quality.
- **S3 (no cheap fix):** action-repeat does not rescue the transition (and modestly worsens it — larger k fails at lower r).
- **S4 (generality):** a second reflex task in a different modality (rotational aiming) collapses onto the same $r \approx 1$ boundary ($r^\star = 0.98$) — the law is not specific to the pit.
- **S5 (a real model through the boundary):** a closed-loop Sonnet agent, judging every frame itself, swept through the transition by injected delay collapses at $r^\star = 1.05$ — and past the boundary its only clears are *delay-rescued misjudgments*, the two failure axes cancelling.

Separately, on the perception axis, a blind vision model *can* judge the jump from a single frame, and obscuring the depth cue (fog) collapses that judgment to chance — at zero delay, i.e. independent of the timing axis.

2 Related work

Five works are closest; we differentiate from each.

- **OmniGameArena** (Lin et al., 2026, arXiv:2606.09826) — has paused (PDQ) vs. latency-controlled real-time (LCRT) clock modes, but only as **aggregate whole-game** comparisons; no ratio transition, no single-decision flip, no action-repeat test. We add all three.
- **VideoGameBench** (Zhang et al., 2025, arXiv:2505.18134) — "Lite" (paused) vs. real-time; names the stale-action mechanism but leaves perception confounded ("it's not just latency"). We resolve exactly that confound by isolating one decision.
- **"Win Fast or Lose Slow"** (Kang et al., NeurIPS 2025, arXiv:2505.19481) — owns the latency–quality tradeoff, but its latency lever is quantization/model-size, which *also* degrades quality, and its ≈200 ms point is an environment action-rate cap, not a task-window threshold. Our $r \approx 1$ collapse varies latency **independently** of the model — the apparatus injects control delay D with the policy fixed.
- **BLINK** (Fu et al., ECCV 2024, arXiv:2404.12390) — canonical "MLLMs fail at depth," but static MCQ. Ours is interactive, parametric, oracle-normalized.
- **Ramstedt & Pal, "Real-Time RL"** (NeurIPS 2019, arXiv:1911.04448) [+ "At Human Speed," arXiv:1810.07286] — the RL parent of the real-time/delay result (monotone degradation for learned policies). We locate an $r \approx 1$ boundary for *pausable VLM agents* and show turn-based deployment crosses it.

We explicitly do **not** claim to be first to study the latency–quality tradeoff (Win Fast) nor first to isolate timing from competence in the aggregate (Real-Time Deadlines, arXiv:2601.13206; OmniGameArena). Our claim is the single-decision flip plus the quality-controlled ratio law, and its generality across a second task.

3 Task and apparatus

A first-person raycaster. The player auto-walks toward a pit; the only decision is *when* to jump. Knobs: pit width (sets the action-window W), fog/visibility (perception), approach speed v, seed. The sim is deterministic and headless; control delay D (ticks) is **injected** — a decision made at tick t is applied at $t+D$ — so D is an independent variable decoupled from wall-clock. Real per-model latency L is measured separately (timed calls) and mapped to the r axis via L/tick. An oracle policy (true pit distance) is the perfect-judgment ceiling.

Ground-truth label. `in_window` = whether a delay-0 takeoff at the decision position clears, computed from the sim's *own* dynamics (not an analytic approximation), so `in_window` exactly predicts `cleared` at D=0. The action window is $W = w_{hi} - w_{lo}$ with w_{lo}, w_{hi} derived by simulating takeoffs; for pit widths beyond the jump reach the window is empty ($W \leq 0$, unclearable) and those configs are excluded from fits (reported, not silently dropped).

$$r = \frac{v \cdot D}{W} \text{ (oracle sweep)} \qquad \text{or} \qquad r = \frac{v \cdot (L/\text{tick})}{W} \text{ (real-model placement)}.$$

4 Results

4.1 Timing law (S1)

Sweeping D for each pit width (each a different W) and pooling against r, the clear/fail outcomes **collapse onto a single transition near** $r \approx 1$; a logistic fit gives $r^\star = 1.17$ (420/1680 empty-window rows excluded). Because the delay grid is discrete, the data bracket the edge rather than pinpoint it: the last clearing configuration lies at $r = 1.04$ and the first failure at $r = 1.30$; $r^\star = 1.17$ is the logistic midpoint of that gap. The transition is razor-sharp because the oracle is deterministic; a stochastic VLM would smooth it into a psychometric curve. The slight offset above 1.0 is physical: a jump firing on the same tick can take off from just past the pit's near edge, extending tolerance marginally beyond the nominal window — hence clears observed at $r = 1.04$.

Real models all land past $r^\star$. Because the decision cost is exactly $r = L/T_{\text{window}}$, where the window is open for $T_{\text{window}} = (W/v) \cdot \text{tick} = 257\,\text{ms}$ at pit width 2 ($W = 1.078$, $v = 0.07$), the $r^\star$ boundary sets a **maximum tolerable decision latency of $r^\star \cdot T_{\text{window}} \approx 300\,\text{ms}$**. We timed five vision models on the identical 20-frame decision battery, in each model's *fastest* single-call configuration ($n = 20$ each):

None clear the 300 ms budget. Even the *fastest hosted frontier model* is ∼4× too slow, and local open VLMs are 30–47×. This is the concrete version of "VLMs are too slow to play games": the whole measured field would need the world **paused, or slowed 4–47×**, to act inside the window — and the gap is a property of the *time regime*, not the model, since r here is L/window with judgment held out entirely. The hosted numbers are a model's best case (this fast config uses no extended thinking; any more deliberate configuration is only slower); local latency is on-device — a real, reproducible closed-loop number.

A closed-loop episode (blocking mode) confirms the local model is non-viable, though the two failure modes should be separated: `llama3.2-vision` also *fails perception* — it answered JUMP on all 20

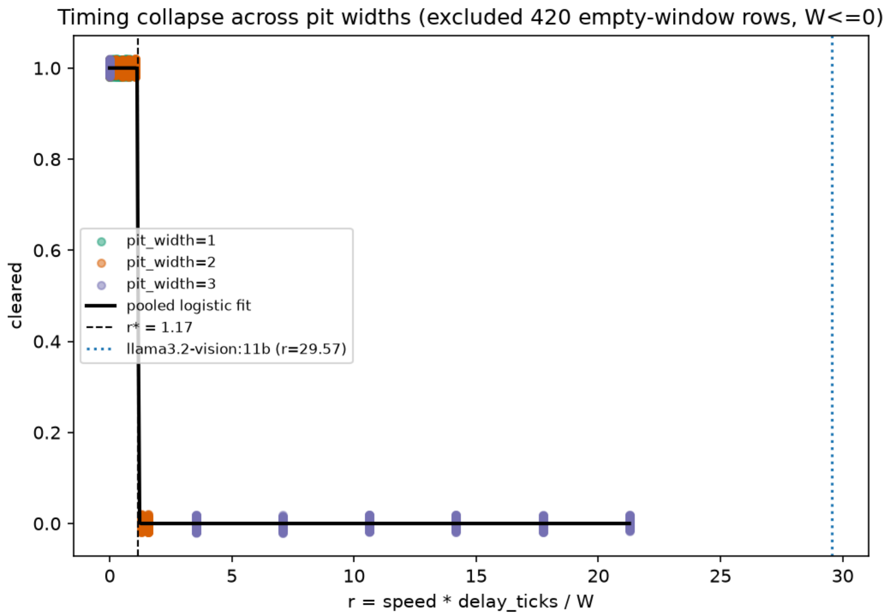

Figure 1: S1. Clear/fail collapses onto a single transition near $r \approx 1$ across pit widths (pooled logistic fit $r^\star = 1.17$).

Table 1: Measured per-model decision latency (median of $n = 20$ single-frame calls, fastest configuration) against the 257 ms window at pit width 2.

Model	Median L	$r = L/\text{window}$
Haiku 4.5 (hosted API)	0.95 s	≈ 4
Sonnet 4.6 (hosted API)	1.45 s	≈ 6
Opus 4.8 (hosted API)	1.53 s	≈ 6
Qwen2.5-VL 7B (local)	7.9 s	≈ 31
llama3.2-vision 11B (local)	12.1 s	≈ 47

frames (no depth discrimination), so in blocking mode it jumps at tick 0 from $x \approx 2.6$ (far from the pit at $x = 10$) and falls. The latency result is the axis-independent point: $r > r^\star$ dooms even a model that judges correctly, the moment the world is allowed to run in real time.

4.2 Same-decision flip (S2)

Restricting to oracle-correct decisions (`in_window` $= 1$), the **identical** decision clears with probability **1.00 at** $D = 0$ and **0.11 once** D **exceeds the window** ($r > 1$). The decision is the same (same `decision_x`, delay-independent); only execution timing changed. This isolates latency from perception directly. (These are unique configs, not raw rows: the oracle emits 60× fog/seed duplicates; the "late" condition is 9 unique configs, 1 of which still clears, giving 0.11.)

4.3 Action-repeat does not rescue (S3)

Holding a decision across k frames does not rescue the timing failure — and in fact modestly **worsens** it. Measured $r^\star$ vs. k (fixed pit width): $k{=}1 \to 1.10$, $k{=}2,4 \to 0.97$, $k{=}8,16 \to 0.71$.

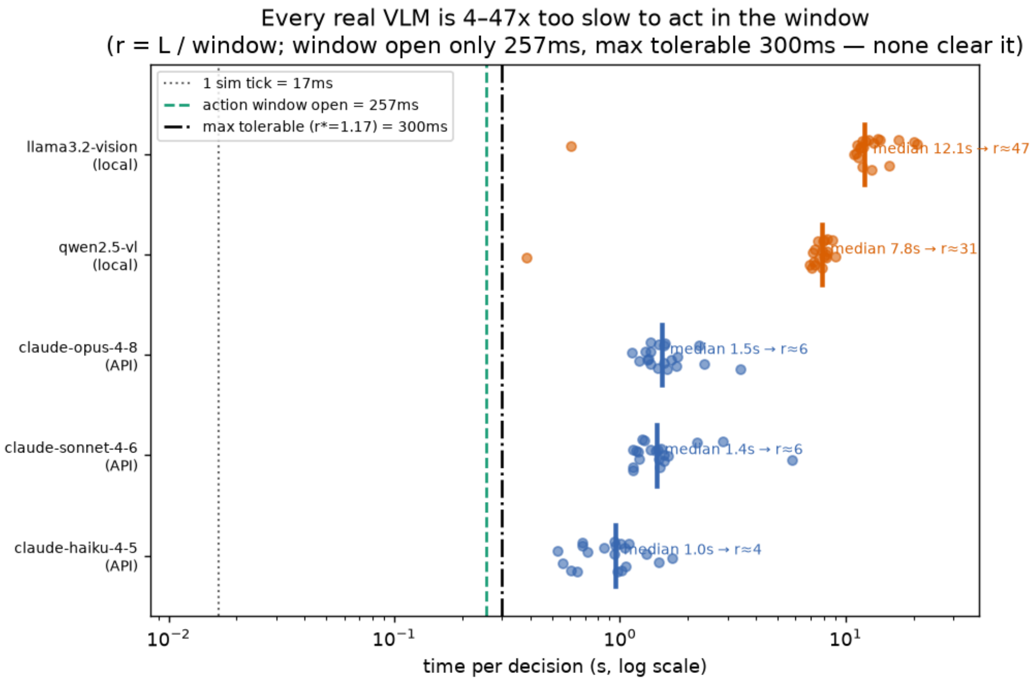

Figure 2: Real per-model decision latency vs. the time budget. Hosted Haiku/Sonnet/Opus ($r \approx 4$–6) and local Qwen/llama3.2-vision ($r \approx 31/47$); all past the $\approx$300 ms max-tolerable budget.

Consulting the policy less often means a decision lands staler (up to $k-1$ ticks later than ideal), so the transition moves to *lower* r (fails earlier), not higher. Action-repeat is a compute/cost lever, not a latency fix. (This sweep is at a single pit width; extending it across widths to confirm the direction is general is future work.)

4.4 Perception axis (dissociation)

A capable vision model *can* judge the jump from a single frame, and fog collapses it. We rendered standing decision frames across approach positions at two visibility levels and had **four blind vision models** — three Claude models (Haiku 4.5, Sonnet, Opus) and one local open VLM (`qwen2.5-vl:7b` via Ollama) — each shown only the image (no coordinates, window, or labels; identical seed-42 shuffled frame set) call JUMP or WAIT on each; we score each call against the ground-truth window. Judgment accuracy at zero delay ($n = 10$ frames per model per condition):

Table 2: Blind single-frame judgment accuracy at zero delay ($n = 10$ frames per model per condition; exact counts shown).

Model	Clear (fog=0)	Foggy (fog=1)	Fog effect
Opus	**90%** (9/10)	50% (5/10, chance)	−**40** pts
Sonnet	80% (8/10)	50% (5/10, chance)	−**30** pts
Qwen2.5-VL 7B (local)	60% (6/10)	60% (6/10)	0
Haiku 4.5	40% (4/10)	40% (4/10)	0

Two things fall out. **(1) With a clear depth cue, judgment scales with model capability**

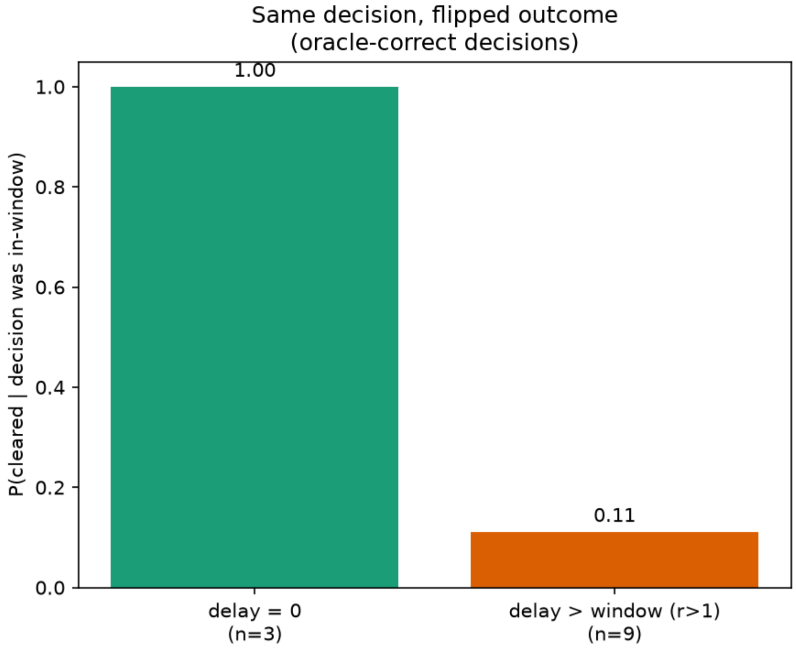

Figure 3: S2. The same correct decision clears 1.00 paused and 0.11 once delay pushes it past the window.

— Opus 90%, Sonnet 80%, Qwen 60%, Haiku 40%. **(2) The fog collapse appears only in the models that actually read depth.** Opus and Sonnet drop to chance under fog, and each **missed every in-window frame** (0/5): they reported no distinguishable pit band even up close, never committed, and would fall. The two weak judges show **no fog effect at all** — but for opposite-looking reasons that are really the same reason: Haiku sits at floor (it fell back on a frame-order heuristic the shuffle nullifies), and Qwen is strongly *JUMP-biased* (it answered JUMP on 18/20 frames, clearing all in-window frames but also most out-of-window ones). Neither is using the depth cue in the first place, so neither has any perception to lose when fog removes it. The fog collapse is a **signature of genuine depth reading**, present exactly where clear-condition accuracy is high. At $n = 10$ frames per cell the per-model contrasts are directional rather than individually significant (Fisher exact, clear vs. fog: Opus $p = 0.14$, Sonnet $p = 0.35$; pooled Opus+Sonnet 17/20 vs. 10/20, $p = 0.04$) — it is the coherence of the pattern across four models, not any single cell, that carries the claim.

This is the perception half of the dissociation: a model that judges well with a clear depth cue drops to chance when fog obscures the pit's distance — **at zero delay**, i.e. independent of the timing axis. Together, perception and timing are separable failure axes on one task. (Caveat: the Claude judges run via a subagent path, not a controlled API benchmark; the local VLM is a real keyless run. Judges are blinded to coordinates and labels but not to the task; the local VLMs' latencies place them on the timing axis at $r \approx 31$ (Qwen) and $r \approx 47$ (llama3.2-vision).)

4.5 Generality (S4)

Is $r \approx 1$ specific to jump-timing, or a property of *any* act-in-a-window reflex? We built a second task in a different modality: a **sweeping-aim** task. The player rotates in place at constant angular

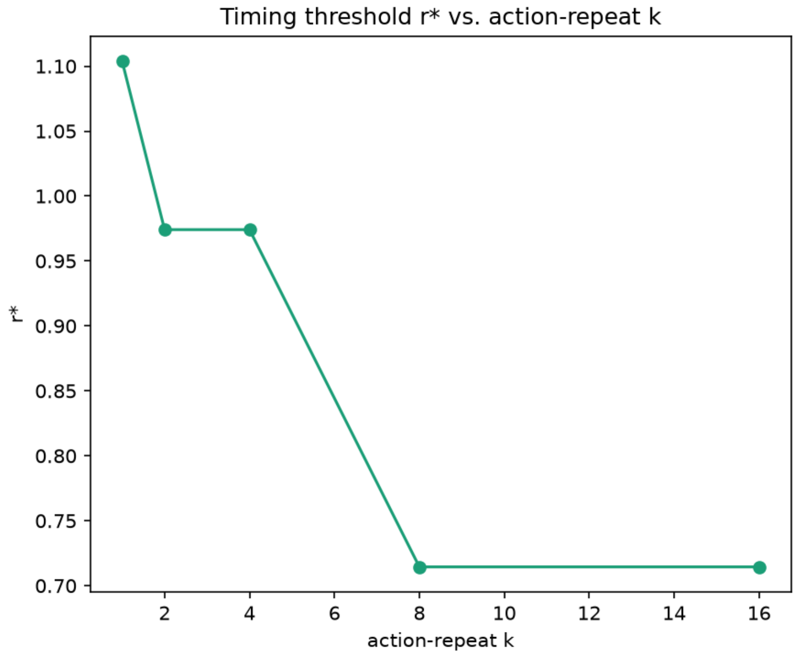

Figure 4: S3. Action-repeat does not rescue the transition; larger k fails at lower r.

speed ω; a target subtends an angular window $[\theta - h, \theta + h]$ (width $2h$); the single decision is *fire*, and a control delay D rotates the aim ωD further before the shot lands — overshoot if it exceeds the window. By construction the dimensionless cost is $r = \omega D/(2h)$, the exact angular analog of the pit's vD/W. The aim episode is pure kinematics and emits the same schema, so the *identical* fit machinery applies.

Sweeping D across five target widths (4.6°–13.8°) and pooling against r, the hit/miss outcomes **collapse onto the same transition near** $r \approx 1$ — logistic fit $r^\star = 0.98$ ($n = 205$). A translational jump task and a rotational aiming task, sharing nothing but the "act inside a window that your delay pushes you out of" structure, land on the **same boundary**. The small offset between the two $r^\star$ (pit 1.17, aim 0.98) is the oracles' firing phase: the pit's takeoff can extend just *past* the near edge (a hair above 1), while the aim fires just *inside* the near edge, forfeiting a sub-tick of tolerance (a hair below 1). The raw step edges show exactly this: the pit's last clear lands at $r = 1.04$ (past the nominal edge), while the aim's last hit is at $r = 0.96$ and its first miss at exactly $r = 1.00$. Both are the same order-unity law. This is the result that makes $r \approx 1$ a *law* rather than a fact about one pit.

4.6 A real model through the boundary (S5)

S1 measures real models' latencies and finds them all far past $r^\star$ ($r \approx 4$–47); the transition itself is mapped by the oracle. Here we close that gap: a real model is swept *through* the boundary with its own judgment in the loop. Because the apparatus decouples sim time from wall-clock, the model's (slow, variable) API latency is free, and r is engineered exactly as in S1: Sonnet 4.6 judges *every grounded frame* of a closed-loop blocking episode (pit width 2, $v = 0.07$, fog 0, seeds 1–6), its committed jump is applied D ticks later, and sweeping $D \in \{0, \ldots, 31\}$ sweeps $r = vD/W$ from 0 to 2, densely around $r = 1$ (72 episodes; 6,612 vision calls).

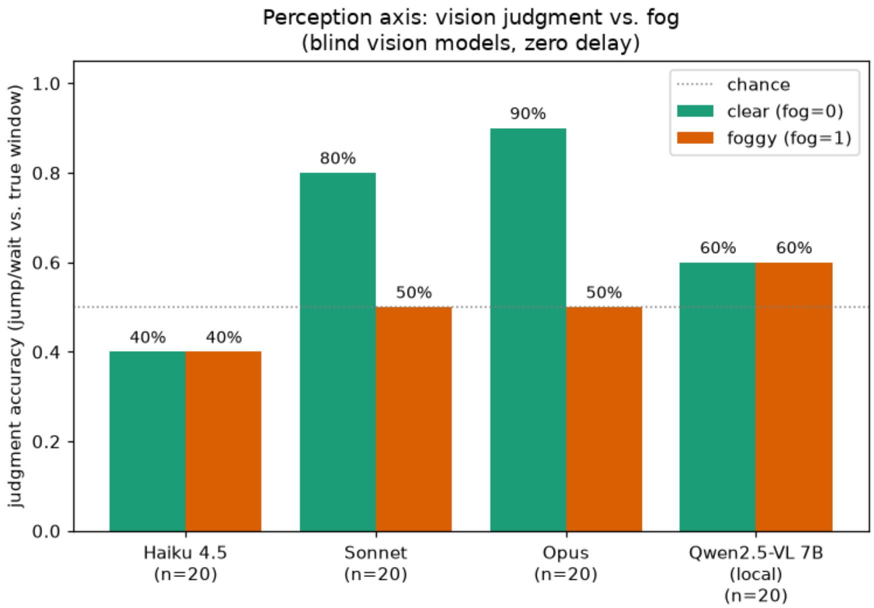

Figure 5: Perception axis. Fog collapses judgment to chance only for the depth-reading models (Opus, Sonnet); JUMP-biased/floor judges show no fog effect.

Judge protocol. The episode prompt used for the S1 latency battery turns out never to elicit JUMP near the window from *any* hosted model: it describes the pit as "the black band across the floor ahead," which stops matching the close-range view — inside the window the pit is a black region filling the bottom of the frame, and the visible checkered floor is the pit's *far* side. The S5 judge instead answers a literal perceptual question ("what touches the bottom border of the image: tiles or black?") at high reasoning effort, with a commit rule of two consecutive JUMP answers. The commit rule matters: in prompt-development batteries we measured isolated false JUMPs at the 1–2% per-call level far from the pit, and any such rate fires almost surely somewhere over the $\sim$90 consulted approach frames — a first-passage failure mode that single-frame accuracy batteries cannot see. We validated the final judge on all 108 approach frames $\times$ 3 samples: zero far-field JUMPs, unanimous JUMP inside the window, and isolated misfires only in the last half-tile before the window ($x = 8.10$–8.24) — the source of the 9/72 early commits below. The bottom-edge cue is nearly ideal in this renderer: the pit's black region first touches the frame's bottom edge at $x = 9.01$, one tick after the window opens ($w_{lo} = 8.92$).

The law holds. Episodes whose decision lands in the window clear **32/32 through** $r = 0.97$, 4/5 at $r = 1.04$, and **0/26 from** $r = 1.10$ **on**; the same logistic fit as S1 gives $r^\star = 1.05$, inside the oracle's bracketing gap $[1.04, 1.30]$ and at its last-clear edge (the same-tick takeoff tolerance of S1). The transition is nearly as sharp as the oracle's, for an unplanned reason: the model's commit position is almost deterministic — 51/72 episodes commit at $x = 8.94$ and 12 at $x = 9.01$, straddling the cue's pixel-level onset (the model begins answering "black" one tick before the black region literally reaches the bottom edge at $x = 9.01$). The anticipated noisy psychometric curve compresses to a near-step; this judge's decision-position variance is sub-tick, so the model behaves as a *slightly-early oracle* rather than a noisy one.

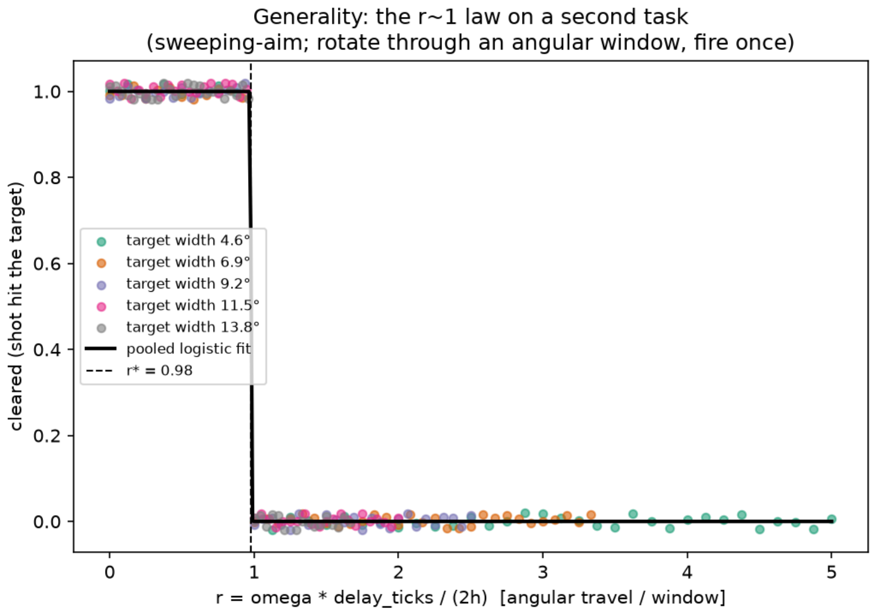

Figure 6: S4. The same $r \approx 1$ collapse on a rotational aiming task ($r^\star = 0.98$, 5 target widths).

Delay-rescued misjudgments. 9/72 episodes commit early ($x = 8.10$–8.24, out of window — a perception error; these are excluded from the timing fit above). At $D \leq 8$ every such episode fails, landing short. In the band $D = 15$–24 ($r = 0.97$–1.56), *every* early episode **clears** (5/5): the delay pushes the too-early takeoff into the window. Past $r \approx 1.1$ the *only* surviving episodes are the misjudged ones; by $r = 2.01$ the delay overshoots even the early takeoff and the rescue band closes. The two failure axes do not merely dissociate — on opposite sides of the boundary they *compensate*, which is only visible because the apparatus labels each decision in- or out-of-window.

5 Discussion

The two failure axes have opposite fixes, and the framing makes the requirement quantitative. A perception failure wants a better model: no amount of extra time helps a judge that reports no distinguishable pit band under fog. A timing failure wants a slower clock: a model is viable exactly when $r = L/T_{\text{window}} < r^\star \approx 1$, so a deployment can cross the boundary by pausing the world ($r \to 0$), slowing it (raising T_{window}), or cutting latency — and the measured field needs 4–47× on the latter two. S5 is the direct demonstration: the same model, same judgment, clears 32/32 below the boundary and 0/26 above it, with nothing varied but r. The practical reading is that pausable or turn-based deployment converts every slow-but-accurate model we measured from failing to viable, with no change to the model; conversely, a real-time benchmark score is a product of judgment *and* time regime, and "VLMs are too slow to play games" is partly a fact about the clock they are handed. S5's delay-rescued misjudgments add a sharper corollary: past the boundary, scores can even *reward* bad perception — the axes compensate, and an aggregate benchmark cannot tell.

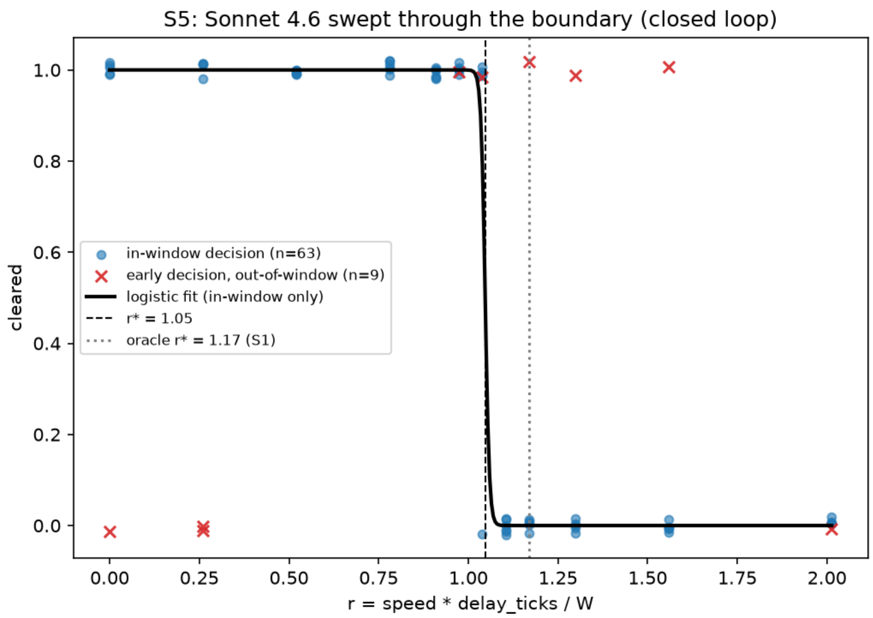

Figure 7: S5. Sonnet 4.6 swept through the boundary in closed loop ($r^\star = 1.05$, in-window decisions). Crosses are out-of-window (early) decisions: they fail below the boundary and are the only survivors past it (delay-rescued perception errors).

6 Limitations

- **The real-model sweep (S5) is one model at one pit width**, and its judge uses a revised prompt (the S1 battery prompt elicits no JUMPs near the window from any hosted model) plus a two-consecutive-JUMP commit rule — a documented apparatus change, not the shipped default. Its r axis is engineered by injected delay with the model's judgment in the loop; the model's own wall-clock latency (∼5 s/call at high effort) remains far past $r^\star$, so a wall-clock-real sweep is still impossible for the measured field — that is S1's point, not a gap. The **latency axis is real**: five models measured directly, all past $r^\star$ (hosted $r \approx 4$–6, local $r \approx 31$–47); network-/hardware-dependent, but real decision latencies, not stand-ins. The anticipated *noisy* psychometric curve did not materialize — this judge's decision variance is sub-tick — so mapping a genuinely noisy transition needs a weaker or more stochastic judge.
- **The perception axis** mixes three Claude judges (run via a subagent path, not a controlled API benchmark) with one real keyless local VLM (`qwen2.5-vl:7b`), at $n = 10$ frames per condition — large enough for the cross-model pattern, too small for per-model significance (Fisher exact $p = 0.14$–0.35). The open VLMs are weak/biased judges — Qwen is JUMP-biased (60% clear, no fog effect) and `llama3.2-vision:11b` is always-JUMP (contributes latency only, not a perception point). A controlled, latency-timed multi-provider API sweep still needs a key.
- **No human baseline** — the oracle stands in as the perfect-judgment ceiling. Human psychophysics is future work.
- **Two tasks, not many.** The $r \approx 1$ law now holds on two tasks in two modalities (translational jump, rotational aim), which is the difference between a fact and a law — but both are single-

decision reflex tasks in the same engine. Multi-decision and richer tasks are untested.

- **Deterministic-duplication caveat:** the oracle emits identical rows across fog/seed; reported counts are unique configs, and the sharp fit reflects zero per-episode noise.

7 Future work

Extending the S5 real-model sweep to more models and pit widths (and to a judge noisy enough to trace a genuinely smooth psychometric curve); a multi-provider latency panel (Gemini/GPT + more open VLMs); a human psychophysics arm; and more reflex tasks (beyond the two here) to test how universal the $r \approx 1$ boundary is.

Reproducibility

Code, data, and figure-generating scripts are at `https://github.com/star7js/judge-vs-react`, under `eval/pit/` (analysis and sweeps) and `raycaster/` (the deterministic sim, including `--task aim` for the second task). The perception judgments (`perception_judgments.jsonl`, 80 rows across 4 models), per-decision latencies (`vlm_latency.jsonl`, 100 rows across 5 models), the oracle and aim sweeps, and the S5 real-model sweep (`eval/pit/realmodel/`: driver, revised judge, 108-frame battery validation, and the 72-episode JSONL) are committed. The local-VLM paths (Ollama) and the Claude subagent judges run without an API key; the hosted-API latency numbers and the S5 sweep required one.

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